

	a max-logit	energy	MSP	entropy	margin
plain cosine head, 20 M	0.626	0.615	0.882	0.899	0.842
ArcFace × z-score, 20 M (A1)	0.907	0.906	0.895	0.905	0.898
ArcFace × z-score, 198 M (A2)	0.830	0.829	0.890	0.881	0.881
B8, 198 M	0.889	n/a	n/a	0.915	0.911
P5, BioCLIP-2 ViT-L	0.916	0.916	0.909	0.916	0.889

open-set AUROC, unseen vs known species (orange: each model's best rule)

